Hawkes Process and REM

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Abstract

This report investigates the connection between Hawkes processes and the Relational Event Models (REMs) to capture self-exciting dynamics in social networks. While standard REMs model event rates using exogenous and endogenous covariates, they typically rely on simple counts of previous events or influence-of-last-event statistics that fail to fully capture the complex temporal structure of self-excitation in human interactions. We try to address this by proposing an approach that injects the self-exciting intensity function of a Hawkes process directly into the REM hazard as covariate. We compare this method against simpler models through simulation, demonstrating its ability to reproduce realistic decay and burst patterns. We further discuss the theoretical and computational limitations of this hybrid approach, particularly regarding the "dilution" of effects in large networks, along with the reasons for not choosing the alternative option, a Multivariate Hawkes Process that implements the social network activity.

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1 Introduction

We can define the Relational Event Models (REMs) as a network model, which allow us to analyze and make inference about the relational processes and dynamics in a social network. These types of models aim to capture how relational events in a network could develop a dependency structure regarding their timing and occurrence. For this reason, we consider as unit of analysis of the REM, a series of tuples (t, s, r), in which s represents the sender, r represents the receiver; while t is the time of the relational event of the edge that connect the sender s to the receiver r. At its core, the REM is defined as a point process for directed pairwise interactions, modeled through a rate function λ. The latter may depend upon sender, receiver, past event history and exogenous covariates. More specifically, the Relational event process, is a marked point process, in which the marks represent the sender and the receiver in a specific event. For this reason, we define a multivariate counting process N, where $N_{s,r}$ is the number of directed interactions between sender s and receiver r. Using some calculations, we can arrive to a general REM definition of the hazard functions $\lambda_{s,r}$ of the relational event $(s \longrightarrow r)$

$$\lambda_{sr}(t) = \mathbb{1}_{\{(s,r)\in R(t)\}}\lambda_0(t)\psi\{\beta(t), x_{sr}(t)\}$$

defined by: the baseline hazard $\lambda_0(t)$, unrelated to the event $(s \longrightarrow r)$, ψ i.e. the relative risk function (in the original formulation of REM of Butts (2008), has been used the exponential risk function); the measurable set of endogenous and exogenous time-varying variables $x_{sr}(t)$ and the effect size β. Also, since in the original formulation the baseline hazard λ_0 is assumed constant, the waiting times between events are exponentially distributed. Furthermore, the covariates can depend on many characteristics, and since the variables could be also endogenous, one may include in one of them the history of the past events (for example the count of the number of past events between (s, r) before time t) as particular endogenous covariate.

Now, given the definition of the REM, if one knows something about the Hawkes process, might already see some similarities. Indeed, we define a Hawkes process as a particular type of inhomogeneous Poisson point process, originally proposed by Hawkes in 1971 as temporal point process, where the event times $t_i \in [0, T]$, such that $t_i < t_{i+1}$, and the right-continuous counting measure $N(A)$, defines the number of events occurring at times $t \in A$. The important characteristic of this model is the self-exciting behavior, which leads to situations where the occurrence of one event increases the likelihood of future events. Hawkes and Oakes in 1974 provided an alternative representation of a self-exciting process as a Poisson branching process. Thinking of each event as a parent, an event occurring at time t_i gives birth to offspring according to a non-stationary process with intensity $g(t - t_i), t_i < t$; these offspring generate their own offspring, and so on...

The influence of the past events is an additive member in the definition of the conditional intensity, so that the total effect of the self-excitation is the sum of the individual effect of each past term. We define the history of all events up to time t as $\mathcal{H}_t$. Indeed, the reason why the process is defined as self-exciting is because the conditional intensity is defined by the history of the process $\mathcal{H}_t$
The conditional density of the process, which quantifies the instantaneous rate of events occurring at time t can be expressed as

$$\lambda(t|\mathcal{H}_t) = \lim_{\Delta t \to 0} \frac{E\left[N \in [t, t + \Delta t) \mid \mathcal{H}_t\right]}{\Delta t}$$

or alternatively, can be defined as

$$\lambda(t|\mathcal{H}_t) = \mu + \int_0^t g(t - u)dN(u) = \mu + \sum_{i:t_i<t} g(t - t_i)$$

where μ represents the background rate of events, and g is the triggering function (intensity function for

the offspring process), which determines the self excitation of the process, since each offspring generated may trigger further events.
So we can say that Hawkes processes, can model both exogenous and endogenous "causes" of events. An exogenous cause happens by the external addition of an event, while endogenous events are self-excited from previous events by a triggering kernel. The form chosen for the triggering function, determines whether the process may depends only on recent history or may have longer term effect. Typically, the function g is monotonically decreasing, so that events occurred further in the past, have a weaker influence to the intensity of the process. One of the options is to take as function g a product of an exponential kernel.

$$g(t, s) = \alpha\beta \exp(-\beta t)$$

α can be interpreted as the overall strength of the excitation, while β is a way to express the temporal decay and so controls the relaxation time of the perturbations induced from past to future events.
It is worth mentioning that the temporal Hawkes process have a closed form solution when we use a background μ constant and as function g for the triggering factor, the exponential kernel:

$$g(t - t_i) = \alpha\beta \exp\left(-\beta(t - t_i)\right)$$

In this case the log-likelihood is simply

$$\ell(\Theta) = \sum_{i=1}^{n} \log \lambda(t_i) - \int_0^T \lambda(t)\, dt$$

and the closed form solution for the integral is just

$$\int_0^T \lambda(t)\, dt = \mu T + \alpha \sum_{i=1}^{n} (1 - \exp\left(-\beta(T - t_i)\right))$$

It could be seen in the R file *Simulate_Hawkes.R* the simulation to actually generate events from a Hawkes process and the inference on the results (thanks to the closed form solution indicated above). The method will be quickly nominated in the Methodology section, where we will also try to find a solution to connect the two processes.

2 Methodology

After having defined the two processes, our goal is to connect them in some way. We will attempt two approaches: injecting into the REMs the self-exciting behaviour that characterizes the Hawkes processes, along with the decay effect that the kernel used present; or do the opposite, i.e. build a Hawkes process, that differentiate in some way the events for each sender and receiver activity (namely the dyads). In this report, and specifically in this section, we will concentrate more on the first method; leaving a small description of the second (and the reason behind the choice of not exploring it in this report) in the Discussion section. The hazard function used for all the three simulation models built is

$$\lambda_{sr} = \lambda_0 \exp(b_1 \cdot x_s(t) + b_2 \cdot x_r(t) + b_3 \cdot x_{sr}(t))$$

where the three covariates are respectively: the covariate related to the events in which the actor s acted as a sender (x_s), that express the activity of the node; the covariate related to the events in which the actor r acted as a receiver (x_r), that express the popularity of the node and at the end x_{sr} which depend on all the times in which the specific dyad $s \longrightarrow r$ happened, to study the reciprocity. Note that for all the three simulation models, the value $b_3 = 0$ for the coefficient of the third covariate. This is due to the choice of not over complicate the simulation models, considering that only the dependence of the first two covariates already give us (especially for the third model) a complicated output for which is difficult

to trace back the effect of the self excitation of the sender or the receiver alone. One could easily add also the effect of the previous (opposite) events to include into the hazard function also the reciprocity effect by simply changing into the code the value of b_3.
All the three models start with the same set up and follow the same logic (obviously with some differences depending on how the covariates are built): we create a network composed of 50 actors, with a defined value for the coefficients b and we start defining the first values of the covariates before entering in the loop: a while loop to repeat the process until we reach the number of events defined, that is for all the models 5000 events. The mechanism behind the simulation of the events use the formula

$$T_{\text{next}} = T_{\text{current}} + \Delta T, \quad \Delta T \sim \text{Exp}\Big(\lambda_{\text{tot}}\Big), \quad \lambda_{\text{tot}} = \sum_i \lambda_i$$

where each λ_i is the hazard function for a specific dyad $s \longrightarrow r$ (i.e. λ_{sr}). After that we sample a dyad (an events between all the possible $N \cdot (N-1) = 2450$ combinations), where each one has a probability of $\frac{\lambda_{sr}}{\lambda_{\text{tot}}}$; update the value of the covariate and recalculate the value of the hazard function λ_{sr}. It is trivial to note that this method is based on simulating an enormous number of hazard functions ($N \cdot (N-1) = 2450$), one for each possible combination (obviously removing the self event like actor 1 $\longrightarrow$ actor 1). The drawbacks of that will be expressed in the Discussion section. Finally, the result is a table that contains the times, the dyad of the sender and the receiver and the values of the covariates for each event simulated.

After having exposed the general procedure of the simulations, we start with the first model. The work has started with a common choice in REM field to implement in the hazard function the history of the process ($\mathcal{H}_t$): use as covariates the count of all the previous events.

$$x_s(s,t) = \sum_{i:\tau_i<t} \mathbb{I}(u_i = s), \quad x_r(r,t) = \sum_{i:\tau_i<t} \mathbb{I}(v_i = r), \quad x_{sr}(s,r,t) = \sum_{i:\tau_i<t} \mathbb{I}(u_i = r \wedge v_i = s)$$

This is the simplest solution to implement the history of all the previous events inside the hazard function. Indeed, in this way each time we have an event that for example include the actor s as sender, we increase the value of all the hazard function that have this specific actor that act as a sender, and the same reasoning also for the receiver. Obviously if we instead consider the reciprocity, we would increase only the value of the specific (reciprocal) hazard function of the dyads just happened, namely if we have the event $s \longrightarrow r$ we would increase the value of the hazard function λ_{rs}. The problem behind this model is that, as we can see in figure 1, in which we analyze the value of a sender (in this case we selected the actor 14 and for coherence we will use the same actor also in the next plots) we can clearly see that using these covariates, we do not consider the decay effect: essentially even if someone has sent 10 messages one year ago, the value of the covariate for that actor will be the same for someone that just sent 10 messages in a row. This clashes with the burst (and decay) effect that characterizes the Hawkes process;

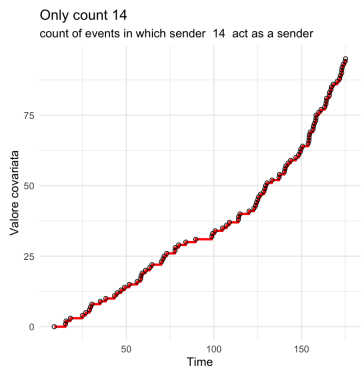

Figure 1: Only count covariate value

and for this reason might seem clever to build a second model that in some way implement the decay effect (someone that sent 10 messages one year ago will have a covariate lower than someone that sent 10 messages now).

The second alternative tries to implement the decay effect by using a second type of covariates, defined as

$$x_s(s,t) = \left(\sum_{i:\tau_i<t} \mathbb{I}(u_i = s) \right) e^{-(t-\tau_s)}, \quad x_r(r,t) = \left(\sum_{i:\tau_i<t} \mathbb{I}(v_i = r) \right) e^{-(t-\tau_r)},$$

$$x_{sr}(s,r,t) = \left(\sum_{i:\tau_i<t} \mathbb{I}(u_i = r \wedge v_i = s) \right) e^{-(t-\tau_{sr})}$$

where τ represents the last time of that event before t. In this case we can see by the formula that we implement a similar version of the decay effect present in the common kernel of the Hawkes process. However, the decay effect is limited to the last event, so that the occurrence of many events one after another does not generate a strong burst effect since we do not sum all the exponential of the negative difference of the previous event. Indeed, we consider the covariate value as a block (the total count of previous events), which decays only with the last event, with a self-exciting behaviour that does not really increase the hazard function thanks to burst of events but only thanks to the total number of events. Furthermore, we have not defined a value of β (as usually done in the Hawkes process to define the speed of the decay effect) and so is fixed to 1, which is definitely too high and so too fast, so that the value of the covariate immediately decays and jumps back in the next event only due to the count part (could be easily seen in the 2, in which we see the difference of the value of the covariate that consider only the total count of previous events and the new covariate just used).

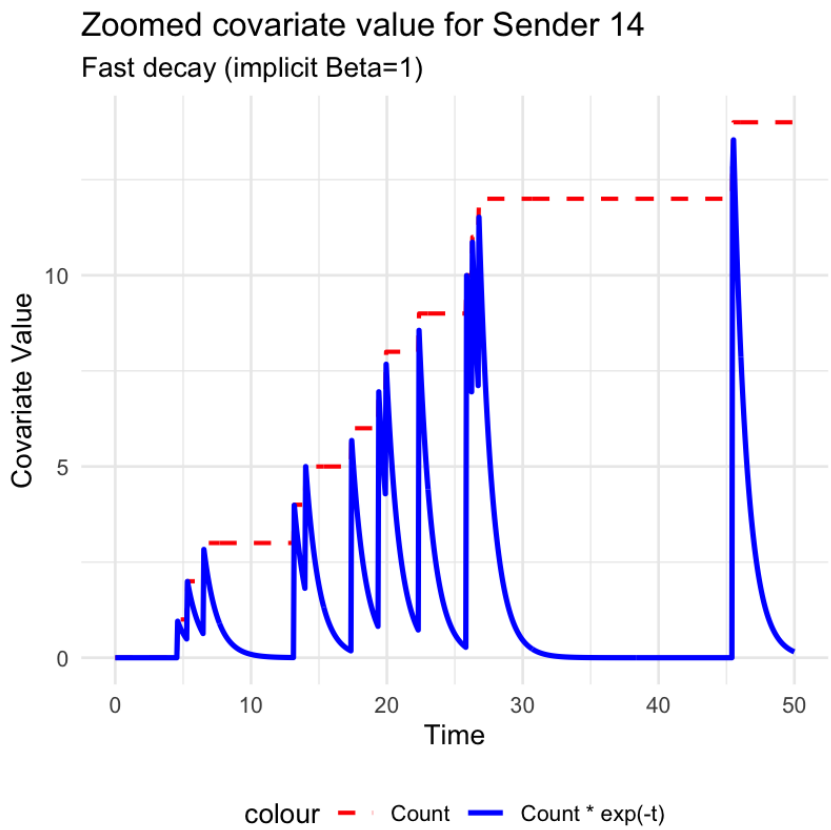

Figure 2: Difference only count and count + decay effect only on last event

The final solution therefore, implement a full Hawkes process for each covariate, leaving out the count of the total events. In this way we can control the decay effect (choosing a value of β) as well as implement in some way the burst of the hazard function that many events one after another generate as in the Hawkes process, by using the sum of the exponential of the difference of all the previous events. The covariates so now are defined as

$$x_s(s,t) = \mu + \sum_{i:\tau_i<t} \mathbb{I}(u_i = s)\, \alpha\beta e^{-\beta(t-\tau_i)}, \quad x_r(r,t) = \mu + \sum_{i:\tau_i<t} \mathbb{I}(v_i = r)\, \alpha\beta e^{-\beta(t-\tau_i)},$$

$$x_{sr}(s,r,t) = \mu + \sum_{i:\tau_i<t} \mathbb{I}(u_i = r \wedge v_i = s)\, \alpha\beta e^{-\beta(t-\tau_i)}$$

This obviously complicate the computation, since now we have to store the history of the time of all the previous events for each specific dyads (for the reciprocity covariate, for the sender and receiver covariates you obviously have the same value of the covariate for each dyad in which a specific actor as as a sender in the first case or as a receiver in the second case). Regarding the computation we have some differences compared to the previous case: since before we have used as covariate the number of previous events, we have had to choose a smaller value of the coefficient b, to avoid the explosion of the exponential. In this case on the other hand, since we have a smaller increment compared to a stable $+1$ for each event, we can (and might) choose a larger value of $b = 0.5$. Regarding the hyperparameters of the Hawkes-process-covariates, we fixed for all of them $\alpha = 0.5$ and $\beta = 0.1$, in which the value of the latter is chosen to guarantees a slow decay to see the effect of the self-excitation last longer.

3 Results

Here, we proceed to analyze the model defined above with Hawkes-process-covariates, to see how the latter change and how the entire model generate time events, due obviously to the change of the hazard function for each dyad.
We can try to reconstruct indeed the Hawkes process that each covariate (in this case for a specific sender) implicitly build in the model. In the 3 we see the result of the intensity (the blue line) of the

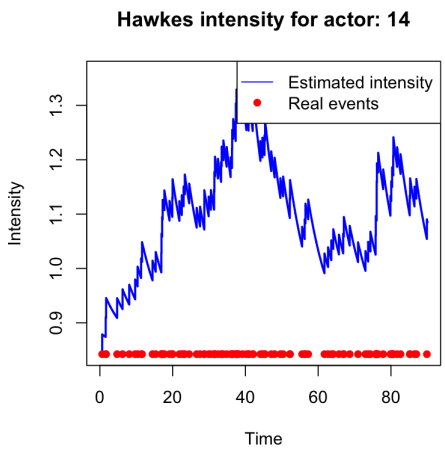

Figure 3: Hawkes process built using MLE for α, β and μ of the covariate of a specific sender

Hawkes process estimated for the events of this specific actor as sender. A crucial caveat must be noted: given the high number of actors and dyads, the sparsity of events leads to significant stochastic variability. Therefore, it is inaccurate to claim that the node's activity strictly follows a Hawkes process with these specific parameters, as the node's behavior is endogenous to the total network history. Rather, we can state that the MLE estimate, given the realized events, yields a trajectory that resembles a Hawkes

process with certain characteristics. However, the estimate of this intensity, is still useful to compare it with the actual value of the covariate of the same node as a sender (see in figure4). In this plot, we

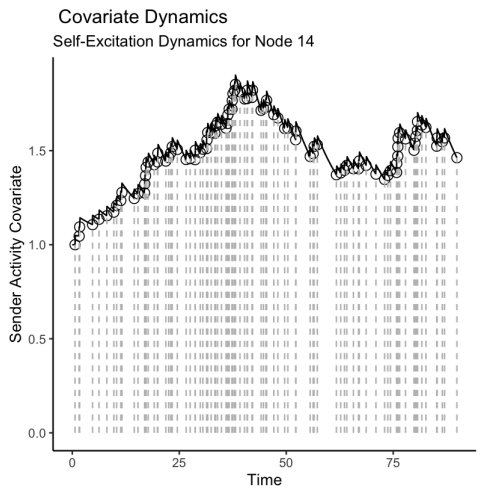

Figure 4: Change of the covariate value according with the events for a specific sender

can see how the two intensities are pretty similar (showing that the MLE procedure has given reliable results), and highlights the self-exciting effect that characterizes the Hawkes process, along with the decay. Indeed, we can see in one of the two plot, how the intensity of the estimated Hawkes process or the value of the covariate in the second case, has reached its peak around time 40. The reason behind that, is to be found in the high number of events that have occurred in succession before that moment. In the plot 4 we see the projection of each event to the covariate value, and before time 40 we clearly see more events projected, that have generated a burst of events that increased the value of the covariate, and subsequently the value of the hazard function for each dyads in which the actor 14 acted as a sender. This effect is indeed the focal point behind the Hawkes process: each event increases the likelihood of that event in the future. On the other hand, around time 60 we clearly see a decay in the value of the covariate, given by the decay effect that also characterize a Hawkes process with exponential kernel for the time: events occurred in the past have less influence compared to events closer, and since around that time we see less events in which the actor 14 acted as a sender, the value of the covariate decreased.

Furthermore, we see in the figure5, the burst effect mentioned before. In this plot we distinguish the

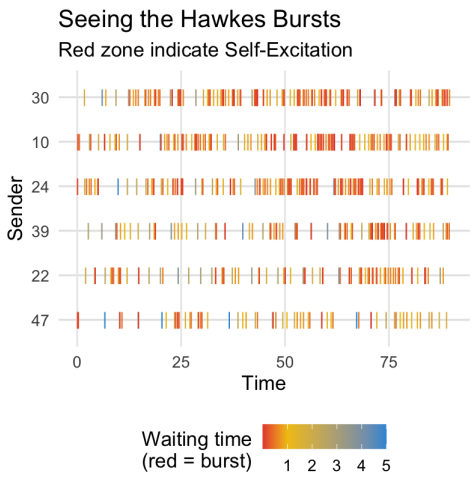

Figure 5: Burst effect in the three most common actor as a sender and three less common actor as a sender

three less and most active actor as a sender; and specifically in the latter we can see the burst effect in the vertical red line: the closer two events are, the redder the second one will be, and since one event in the self-exciting field increase the likelihood of future (same) events, we can expect group of vertical lines (events) with a red color separated by moments with less events (decay effect) and next events indicated with a blue line. Obviously this procedure will be more evident in the actor that has been sender frequently compared to actor less active as a sender. The reason, as already said, lies in the fact that the burst effect lead that specific actor to act as a sender more time in the network.

The burst effect could also be seen in the slope of the count of events for the actors. Indeed, in the

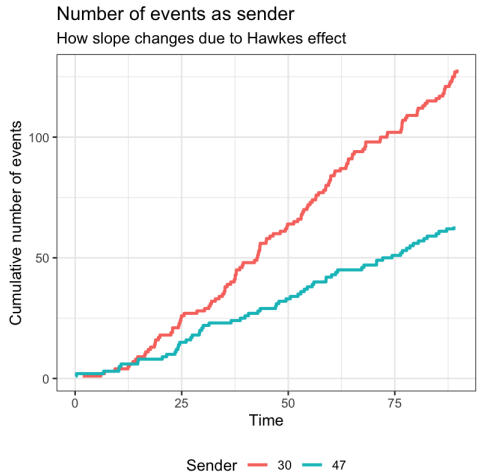

Figure 6: Inclination of the total number of events for most and less actor as a sender

6 we see how the actor more active as sender (actor 30) compared to the less active (actor 47) have an higher slope, and the reason lies in the moments (like around time 30) in which while the first one continuously increase the number of events in which acted as sender, while the second one does not appear as a sender in the network anymore, leaving the count line flat.

The last plot, show instead the difference between a covariate that only count the number of events compares to the one that take into account the self-exciting effect. Note that to create the only-count

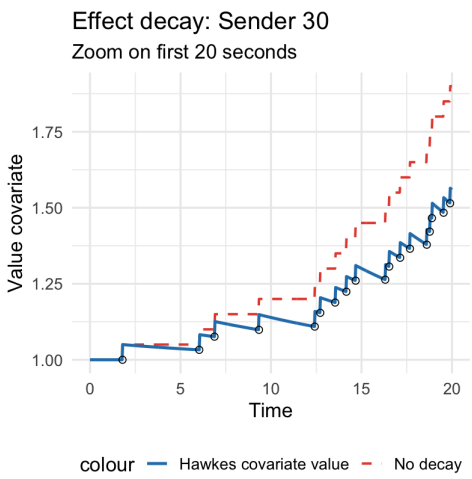

Figure 7: Difference with an only-count covariate

line in 7 we do not add 1 for each event but $+\alpha\beta$, this to maintain the order of magnitude of the covariate

compared to the one generated in the simulation. This plot shows the decay effect that as mentioned, was not present in the first covariante used: the one that have implemented history of the process using only the count of events. Furthermore, compared to the covariate mentioned before, we avoid the risk of explosion, since the value of the covariate does not increase continuously, together with the hazard function for a specific dyad and so not leading to a network where only one event happens. On the other hand, the Hawkes-covariate, have as drawback the fact that could never reach the peak that a normal Hawkes process usually have. Specifically this last point will be discussed in the next section.

4 Discussion

The results seems to show that treating the Hawkes intensity as a dynamic covariate is a valid way to inject memory into REM. Unlike the simple model with as covariate the number of previous events with that characteristic, which implies memory never fades, or the second model, which only cares about the last event; this approach integrates the entire recent history weighted by time, exactly as in the Hawkes process. However, this is only the first alternative to try to find some connection between the REM and the Hawkes process, and still presents some drawbacks.
First of all, as we can deduct by the images above, we have an increase of the hazard function at each event (since the covariate increase), that however does not compare to a traditional Hawkes process. Indeed, in this case the hazard function is not composed only by the Hawkes process, but use the latter just as covariate inside the exponential. The problem is that compared to a traditional Hawkes process, where we directly add the self-exciting effect, here many events one after another increase the value of the covariate (that is a sum of exponentials), which is not an additive member in the hazard function of that dyads, but a multiplicative one. The results at the end is that the self-exciting effect is "diluted" so that the hazard of all the possible dyads is almost the same for all the possible $s \longrightarrow r$ events. This problem is accentuated by the high number of actors in the network. Indeed, even if the actors are "only" 50, the total possible combinations gives us $N \cdot (N-1) = 2450$ hazard function λ_{sr}, each one with probability at the start of $\frac{\lambda_{sr}}{\lambda_{\text{tot}}}$, and this means that each event is more likely to happen if the probability is higher, while in the Hawkes process the increase of the likelihood directly touch the intensity of that event. We

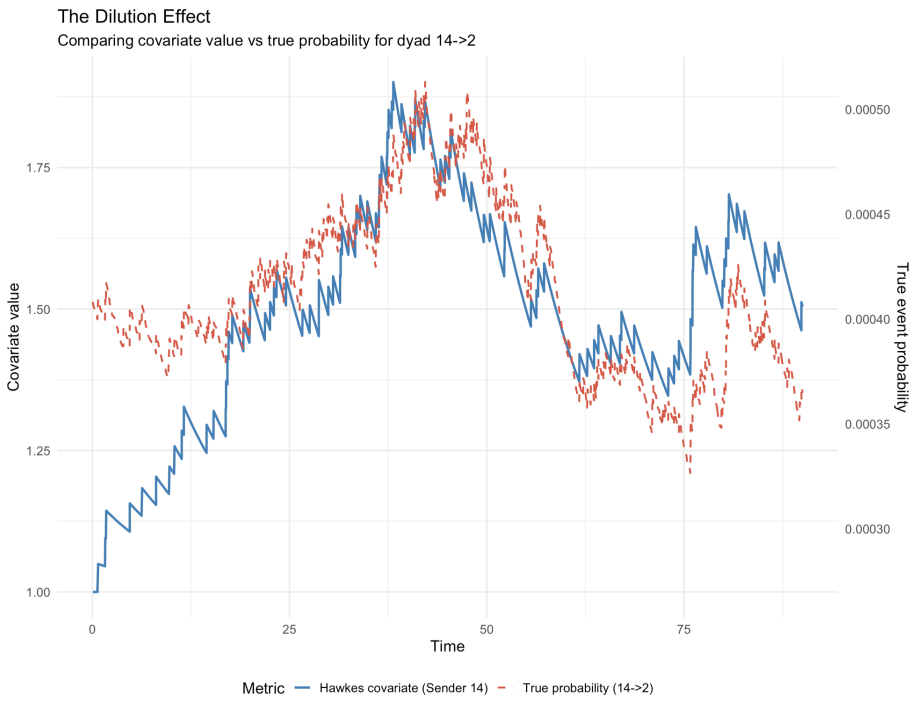

Figure 8: Comparison between covariate and specific probability for a dyad

can see this in 8, where we highlight this limitation of injecting Hawkes process into a large network REM. The blue line shows that the covariate (x_s) behaves as intended, capturing the self-exciting bursts where the intensity nearly doubles (from ≈ 1.0 to ≈ 1.9) around $t = 40$.
However, the dashed red line reveals that this strong effect is effectively diluted by the network size. Due to the competitive nature of the hazard function in a network with $N(N-1) = 2450$ dyads, the actual probability of the specific event $14 \to 2$ remains extremely low, oscillating only between 0.00035 and 0.00050. This demonstrates that even with a strong self-exciting and decay effect at the actor level, the results lead to a barely visible increase in the actual event likelihood. Consequently, while the model correctly encodes the memory of past events, its impact on the generative process is heavily diluted, suggesting that in large networks (even if the dimension of $N = 50$ is not so large for a common network), the self-exciting mechanism may be insufficient to drive significant changes in the dynamics between the actors.

On the other hand, if we increase too much the influence of the self-exciting effect (maybe choosing a value for the coefficients b higher) we fall into the opposite problem: since we have the exponential of the linear part, we might arrive to an increase of the hazard function so big, that all the other events become essentially impossible and the event that the model simulates is always the same. The problem behind that tradeoff is to be found in the mathematical formulation of the model: while in the Hawkes process the sum of exponential of difference in time events is added to the intensity of the model, here the sum is inside an exponential (due to the formula of the REM). This means that, as said before, the effect is no more additive but multiplicative, leading to either violent explosion (with normal coefficients b) or diluted effect (as shown above). Finally, we also have the computational problem: to keep the history of the process we basically have a list of lists, while in the Hawkes process since the update is recursive, all the computation is less expensive.

We have however the other option as stated at the beginning: instead of inject the Hawkes inside the REM we can try to implement the network dynamics inside a Hawkes process. To implement this idea, the straightforward idea is to build a multivariate Hawkes process, that is defined to count both the self excitation between a single event (how the event $s \longrightarrow r$ increase the likelihood of future $s \longrightarrow r$ events) and the cross excitation (how the event $s \longrightarrow r$ increase the likelihood of future different type of events like $p \longrightarrow m$). Fundamentally what we are saying is that each possible dyad in the network is a different event that the multivariate Hawkes process might take into account to update the different intensity function. The previous solution is based on the fact that in a closed system like the REM, where the sum of probabilities across all dyads must normalize to 1, local self-excitation is effectively suffocated by global noise. Conversely, a true multivariate Hawkes process does not suffer from this constraint: it models absolute event rates, rather than relative dyadic choices, allowing a specific node to explode in activity without necessarily "stealing" probability mass from the rest of the network. The problem is that, even if a multivariate Hawkes process seems to offers a natural framework for modeling self-excitation and cross excitation in temporal point processes, their direct application to complex social networks presents significant challenges.

First of all, a multivariate Hawkes process defines the conditional intensity for a directed event from node s to node r as:

$$\lambda_{s,r}(t) = \mu_{s,r} + \sum_{(x,y)\in\mathcal{E}} \sum_{t_k^{(x,y)}<t} \alpha_{(x,y)\to(s,r)} \phi(t - t_k^{(x,y)})$$

where $\alpha_{(x,y)\to(s,r)}$ represents the influence of a past event of dyad (x, y) on the future occurrence of an event of dyad (s, r). We have already said that in the case of our simulation, this implies 2450 possible dyads. A fully specified multivariate Hawkes process would require estimating an influence matrix of size 2450×2450, resulting in over 6 million parameters. This makes the estimating part not only computationally unsustainable, but also statistically unidentified without imposing restrictive

constraints (e.g. assuming a mean-field approximation, in which we estimate that on average everyone influences each other in the same way), which paradoxically make the model computationally feasible, but not noteworthy for a network analysis.
Furthermore, a core requirement of a network analysis is to estimate the effect of exogenous attributes and endogenous temporal dynamics. Trying to incorporate exogenous covariates into the intensity of the model, might lead analytical difficulties:

- Positivity Constraints: unlike the multiplicative hazard of a REM ($\lambda = \exp(\beta^T X)$), the additive form of a multivariate Hawkes process requires that the sum of components remains positive. This makes the inclusion of covariates that might have negative effects for a specific events complicated.
- Model Limitation: using a "simple" multivariate Hawkes process would also force the analysis to rely almost exclusively on the history of the model, ignoring other valuable information in the network model and so limiting the analysis that a social network model assumes.

5 Conclusion

In this project, we explored the intersection of two similar processes to analyze time data: the Relational Event Model (REM) and the Hawkes Process. Our primary objective was to bridge the gap between the structural network definition of REM and the temporal self-exciting effect of the Hawkes processes. Our solution is based on a construction of dynamic covariates that mimics the Hawkes intensity formula. This was done with the objective to import the memory mechanism of the Hawkes process into the linear predictor of the REM.
Our results confirm that this "injection" method is superior to covariates defined by simple count, that lack the temporal decay characteristic, and single-event memory models, which fail to capture cumulative bursts. The visual analysis of Hawkes bursts and covariates dynamics demonstrates that the proposed model shows both the self exciting effect and the decay effect that characterizes these events.
However, this hybrid approach present also some problems. We identified a fundamental trade-off between the "dilution" of the self-exciting effect after the occurrence of one event (the effect of a single burst is moderated by the competitive noise of the entire list of other hazard function, one for each dyad) and the risk of exponential explosion (with as consequences the network monopoly by a single event). Furthermore, we quickly mentioned the second alternative, with some drawbacks that lead us to the exclusion of that option. Future work should focus on optimizing the second alternative, along with a deeper exploration of the field to find a possible way to define the REM as Hawkes process, or viceversa (maybe defining a new covariate that may avoid the complexity of the entire Hawkes process, but that could capture all the information and effects that a traditional Hawkes process have). Furthermore, future analysis could remove the inclusion of the Hawkes background rate μ directly inside the covariate, to avoid mathematical redundancy with the REM's baseline hazard λ_0. The inclusion of the redundant parameter in our simulation was maintained to guarantee the stability of the model.

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